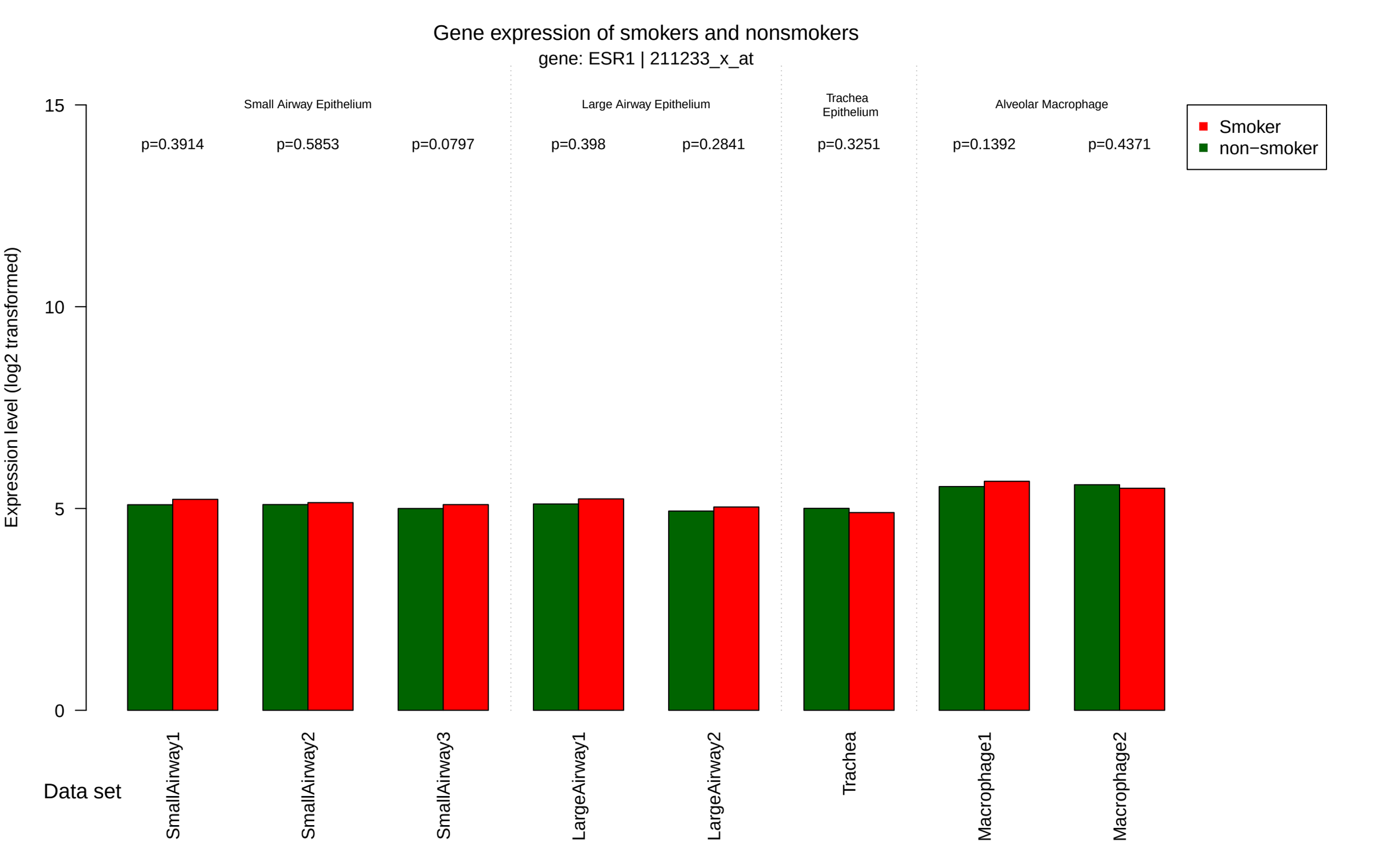
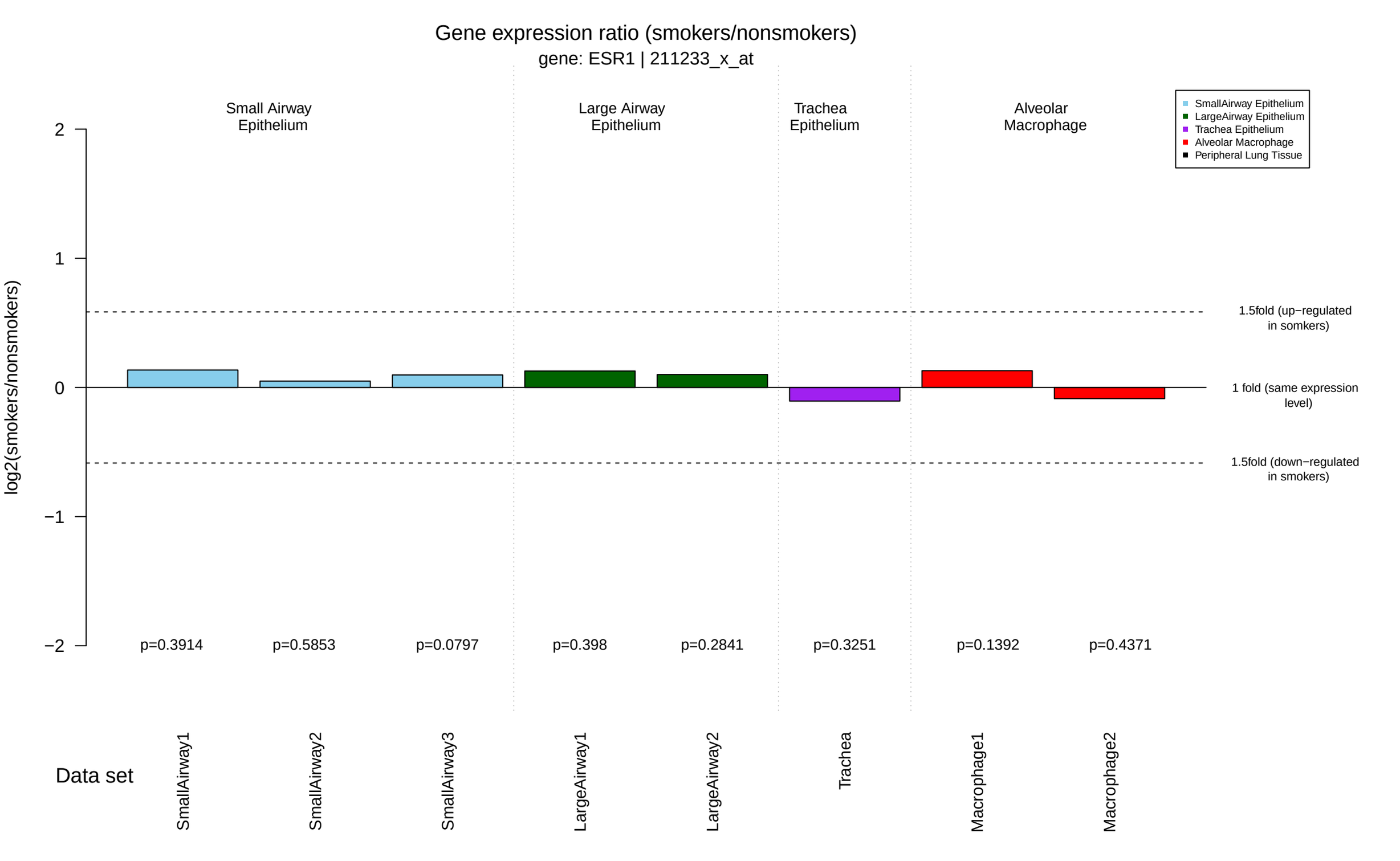
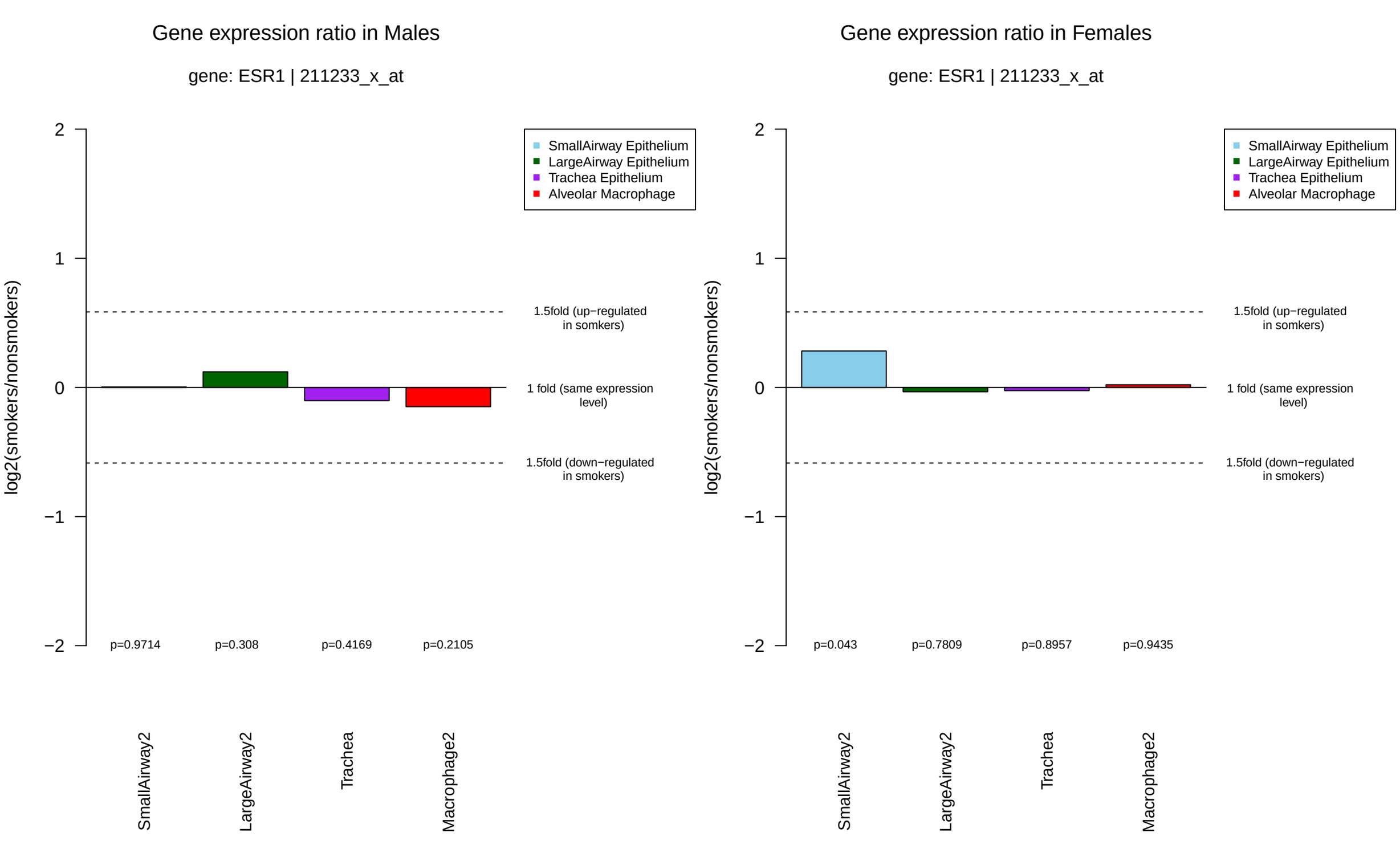
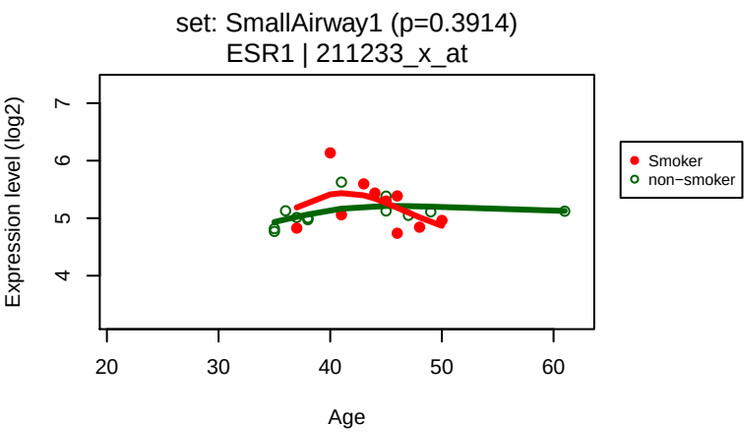
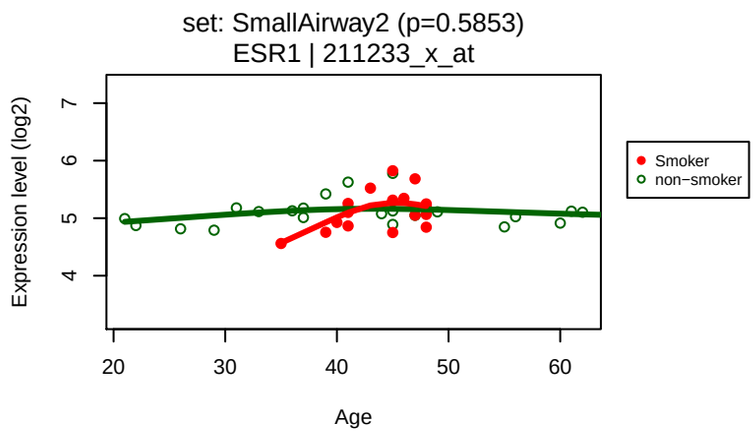
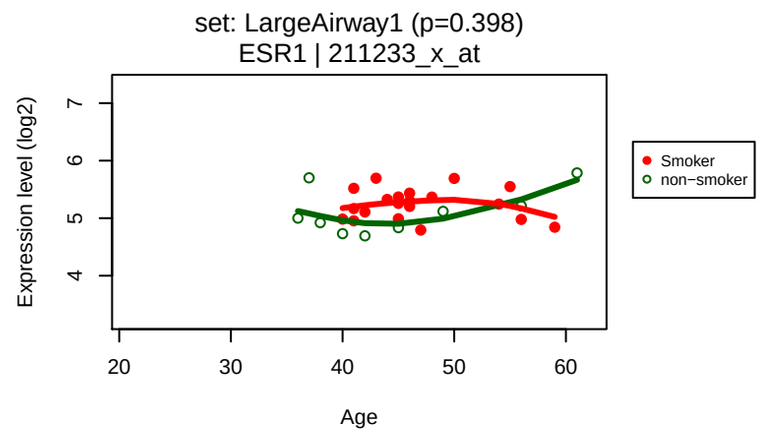

Gene expression of smokers and nonsmokers
gene: ESR1 | 211233_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3914
p=0.5853
p=0.0797
p=0.398
p=0.2841
p=0.3251
p=0.1392
p=0.4371
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

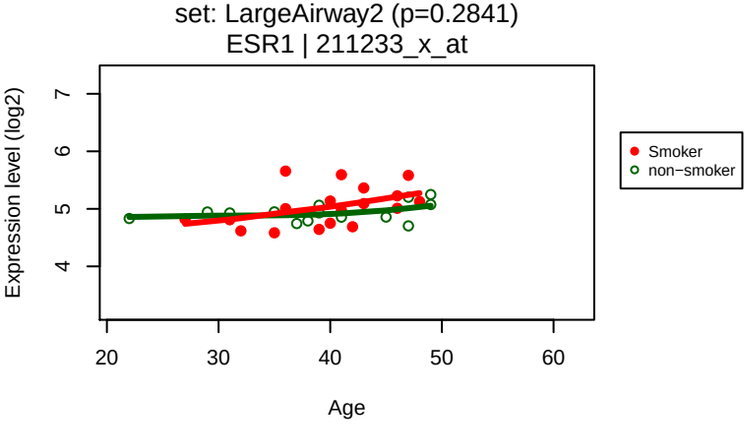

Gene expression ratio (smokers/nonsmokers)
gene: ESR1 | 211233_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3914
p=0.5853
p=0.0797
p=0.398
p=0.2841
p=0.3251
p=0.1392
p=0.4371
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

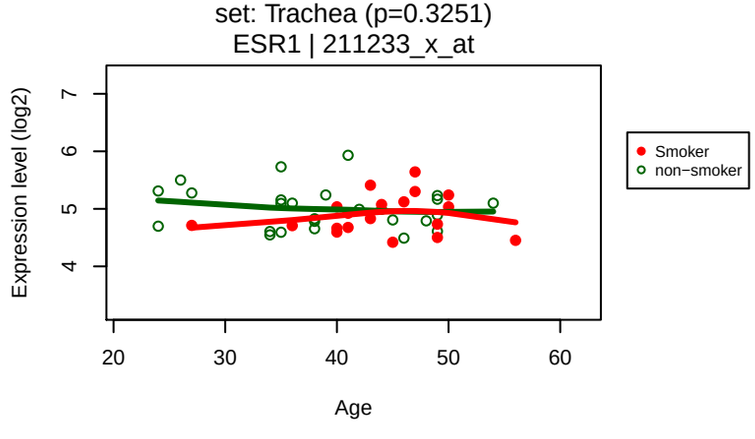

Gene expression ratio in Males
gene: ESR1 | 211233_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9714
p=0.308
p=0.4169
p=0.2105
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ESR1 | 211233_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.043
p=0.7809
p=0.8957
p=0.9435
SmallAirway2
LargeAirway2
Trachea
Macrophage2

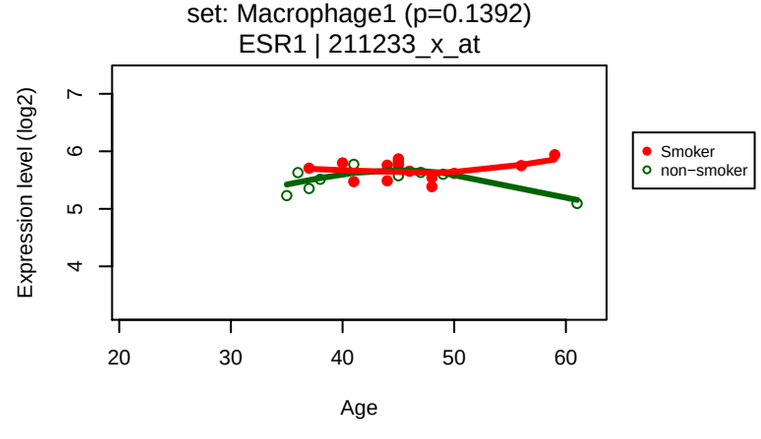

set: SmallAirway1 (p=0.3914)
ESR1 | 211233_x_at
Expression level (log2)
Age
Smoker
non−smoker

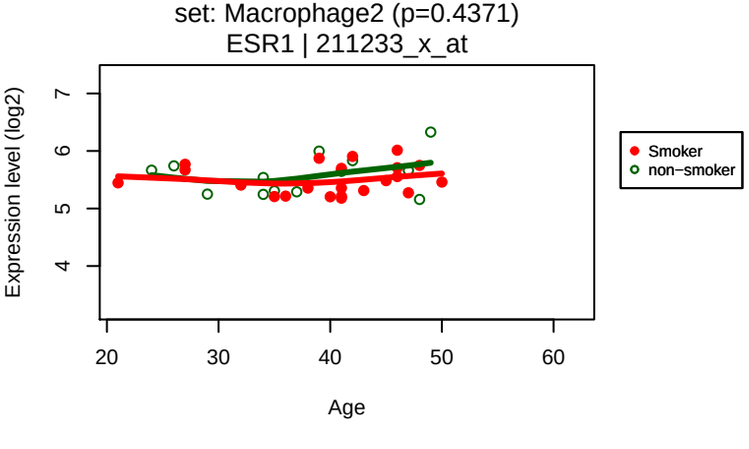

set: SmallAirway2 (p=0.5853)
ESR1 | 211233_x_at
Expression level (log2)
Age
Smoker
non−smoker

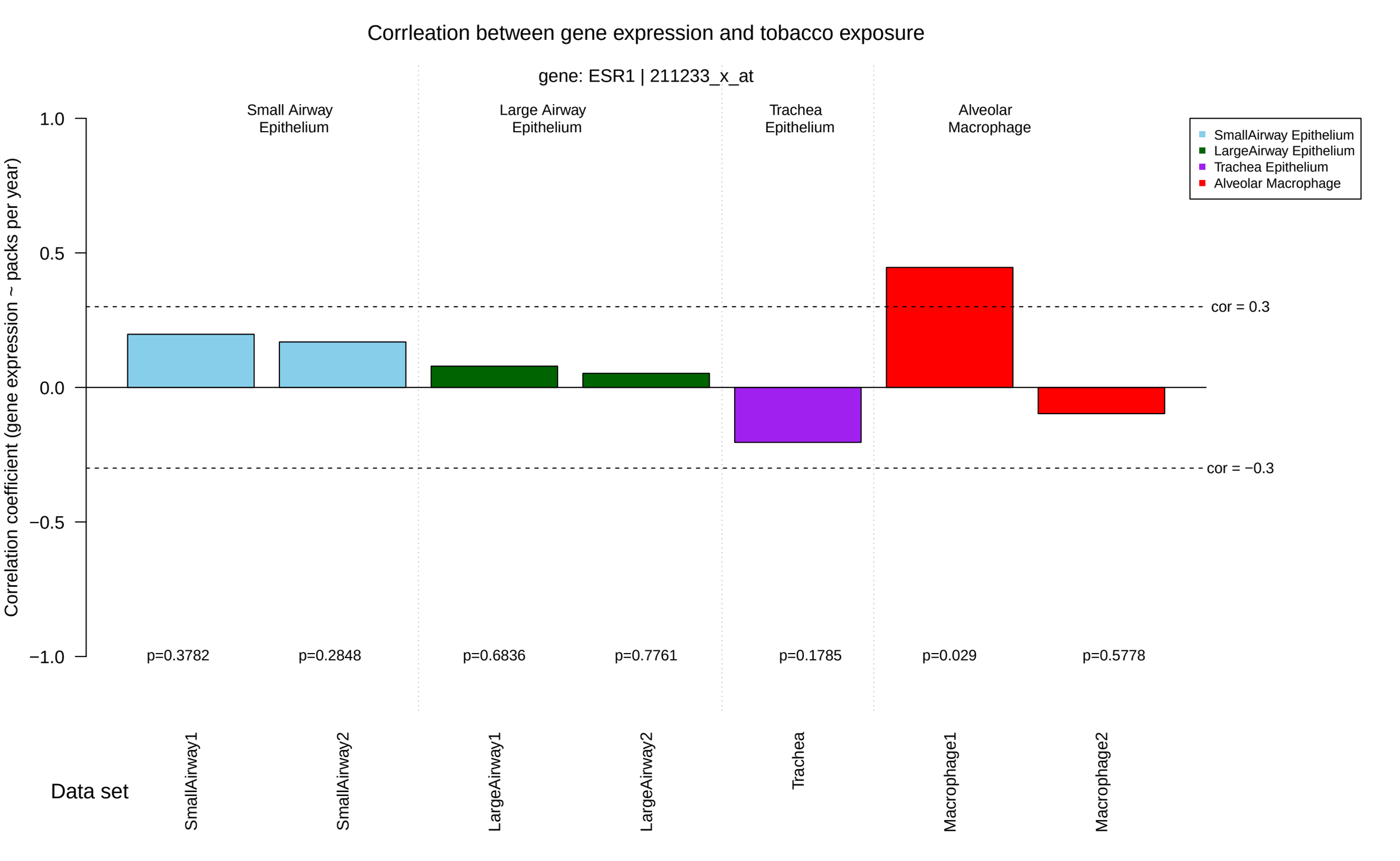

set: LargeAirway1 (p=0.398)
ESR1 | 211233_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2841)
ESR1 | 211233_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3251)
ESR1 | 211233_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1392)
ESR1 | 211233_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4371)
ESR1 | 211233_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ESR1 | 211233_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3782
p=0.2848
p=0.6836
p=0.7761
p=0.1785
p=0.029
p=0.5778
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2